SCHEMAS

AND ETL/ELT

IN THIS PRESENTATION

- What is Data Modeling?
- Data Modeling Levels
- Dimensional Modeling
- Star Schema
- Snowflake Schema
- Slowly Changing Dimensions(SCD)
- SCD Types
- ETL
- ELT
- ETL vs ELT

WHAT IS DATA MODELING?

- Data modeling ensures that data is understandable, scalable, and reliable
- The process of organizing data structures
- Poor data modeling leads to slow queries and incorrect insights
  - Major impact for businesses!
- Defining tables, relationships constraints
- Supports consistency and performance
- Foundation for analytics and reporting

DATA MODELING LEVELS

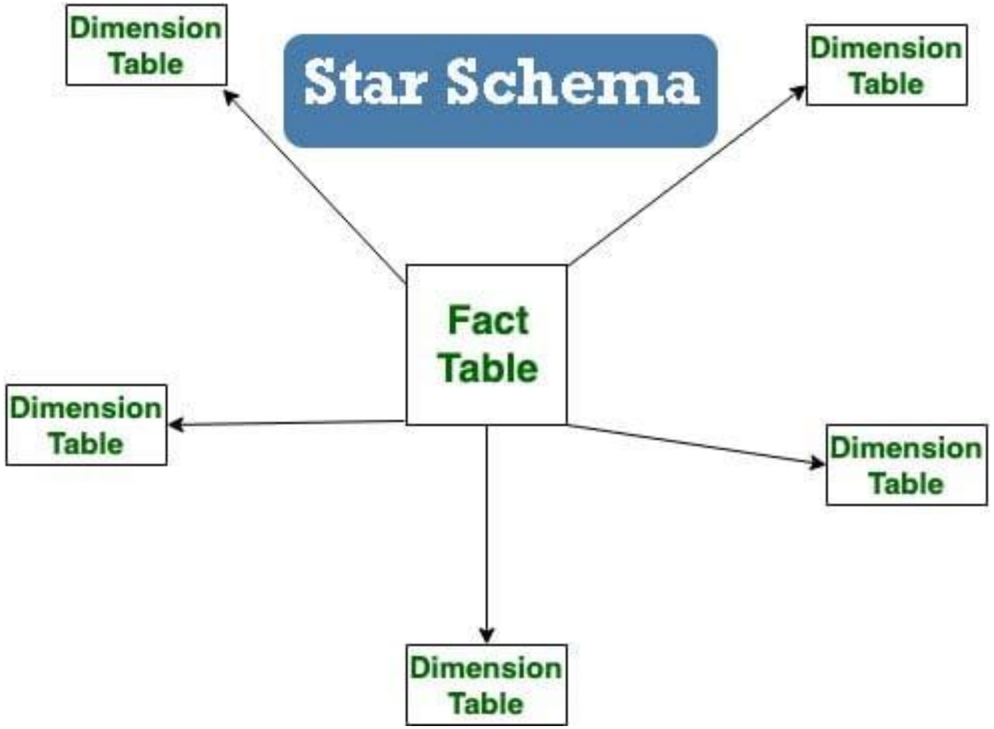
- Data modeling happens in layers/levels
- Conceptual model – the high-level business view
  - What data exists?
- Logical model – the detailed structure without technology
  - How is the data related?
- Physical model – the implementation-specific design
  - How is it stored?

DIMENSIONAL MODELING

- Dimensional modeling is one of the data modeling techniques used in data warehouse design
- The main goal is to improve data retrieval, so it is optimized for SELECT operations
- Used by many OLAP systems
- Elements of dimensional model:
  - Facts – measurable data elements that represent business metrics of interest
  - Dimensions – descriptive data elements that are used to categorize or classify data
  - Attributes – characteristics are used to filter, search facts, etc.
  - Fact Table – the central table that contains the measures or metrics of interest, surrounded by dimension tables
  - Dimension Table – de-normalized tables that describe attributes of metrics, joined to fact table by foreign key

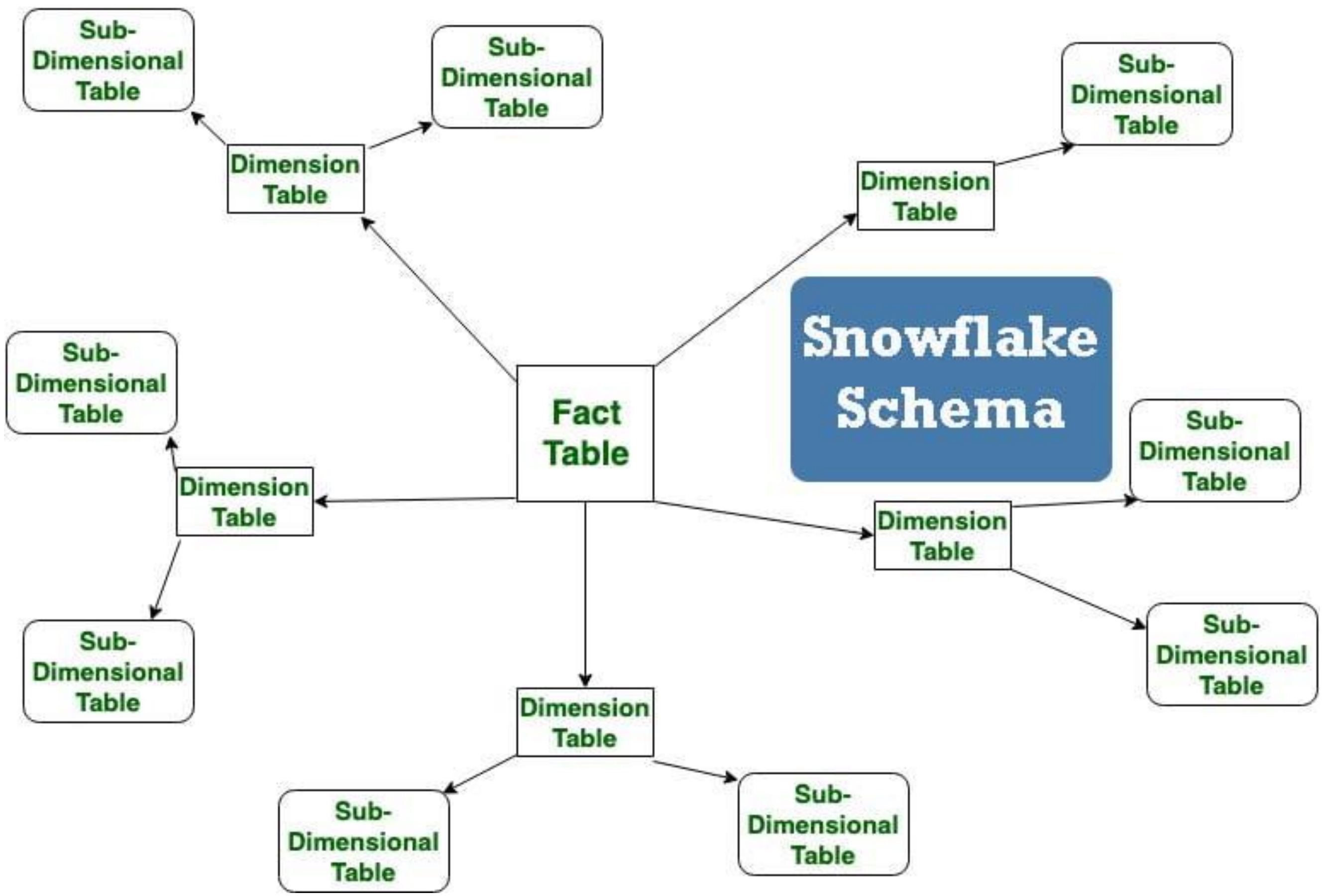
STAR SCHEMA

- Star Schema is the most common dimensional model
- Organizes data into a central fact table linked to multiple dimension tables
- Makes analytical queries fast, simple, and efficient
- Disadvantages include redundancy, limited flexibility, and weak many-to-many support



SNOWFLAKE SCHEMA

- A technique where dimension tables are normalized into multiple related sub-tables
- An extension of star schema, designed to handle complex hierarchies and reduce redundancy

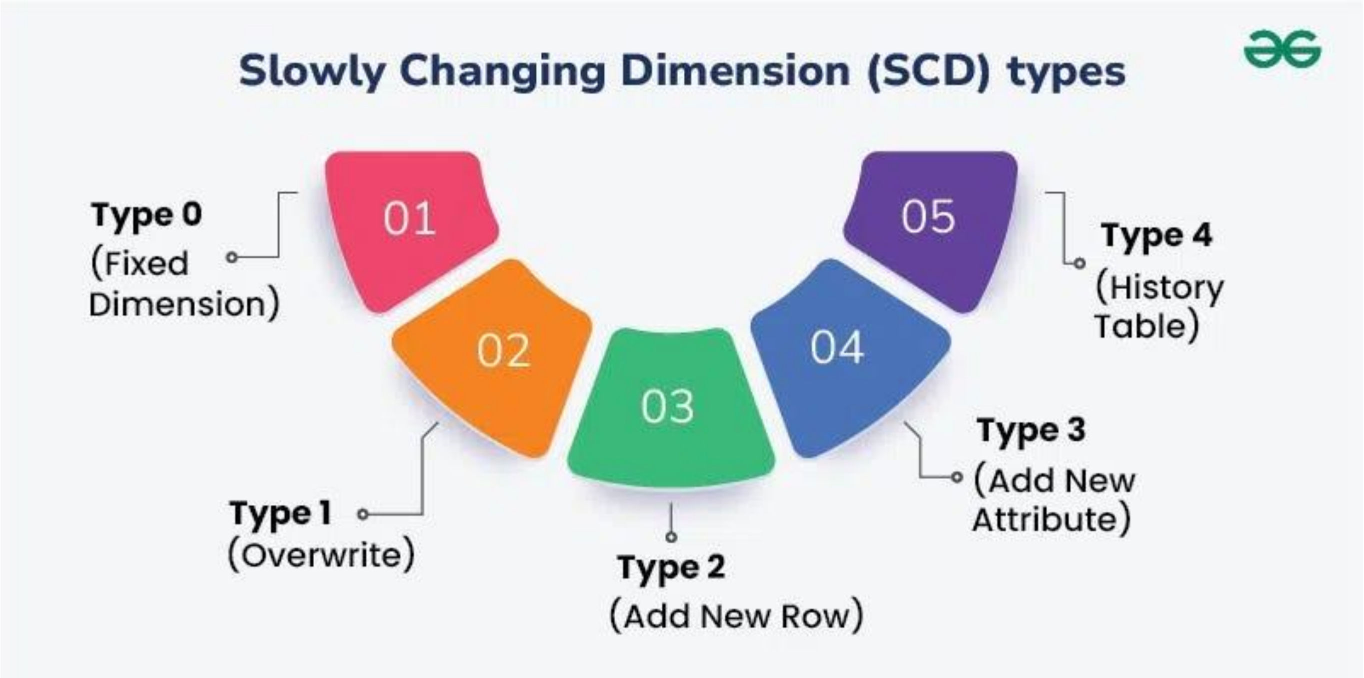


SLOWLY CHANGING DIMENSIONS (SCD)

- Slowly Changing Dimensions (SCD) is a concept that refers to a dimension that stores data which may change over time, slowly and often unpredictably
- SCD's record and preserve past alterations in data for analysis, reporting, and decision making
  - Helps keep data current and useful for future use
- SCD's define how changes are handled

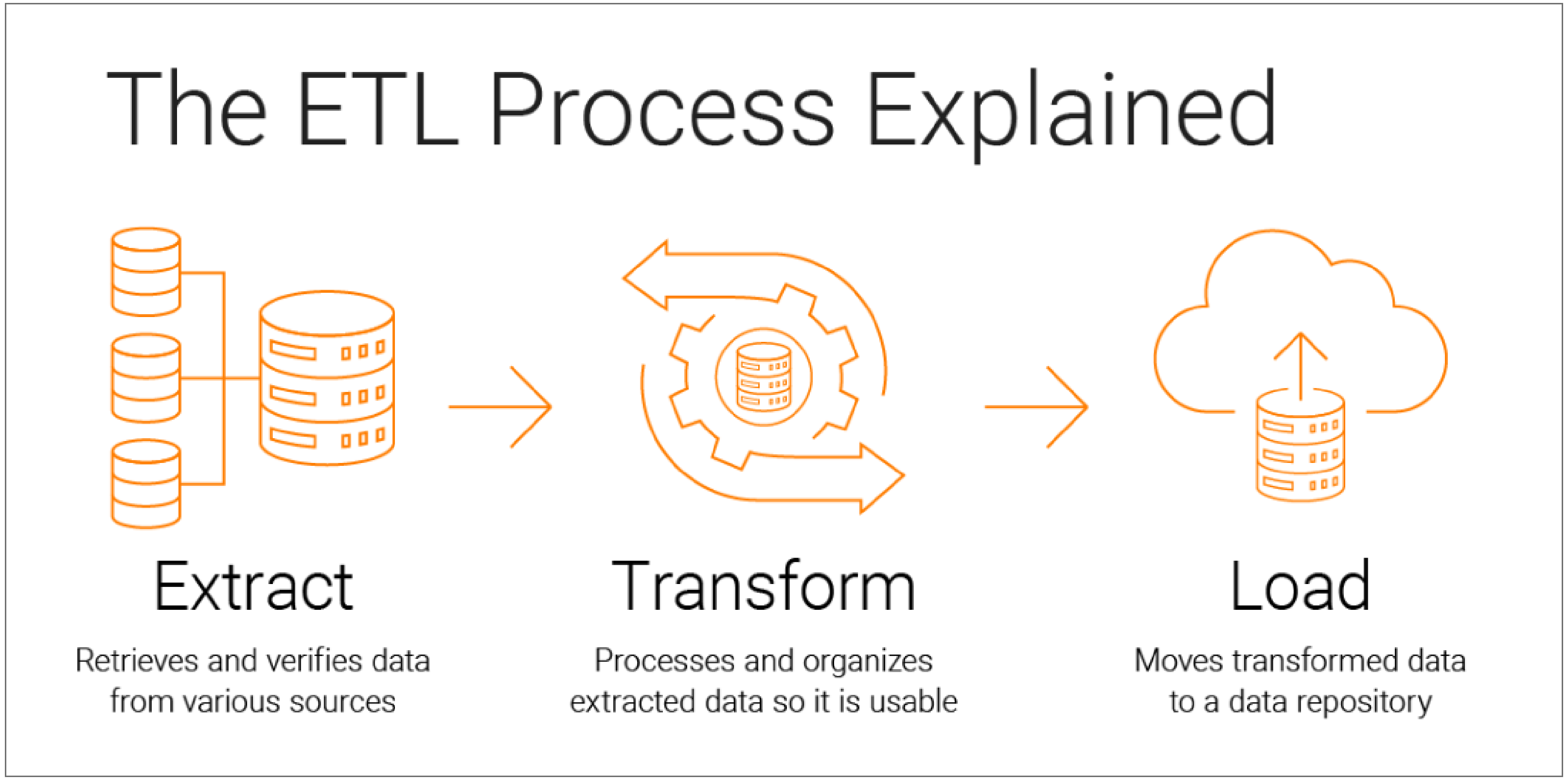
SCD TYPES

- **Type 0**: Fixed Dimension allows no changes
- **Type 1**: Overwrite history
  - Good for correcting errors
- **Type 2**: Add new row every time there is a change, but history is retained
- **Type 3**: Add new attribute records changes by adding an attribute that will hold the previous value
- **Type 4:** History table holds all past information from the current table



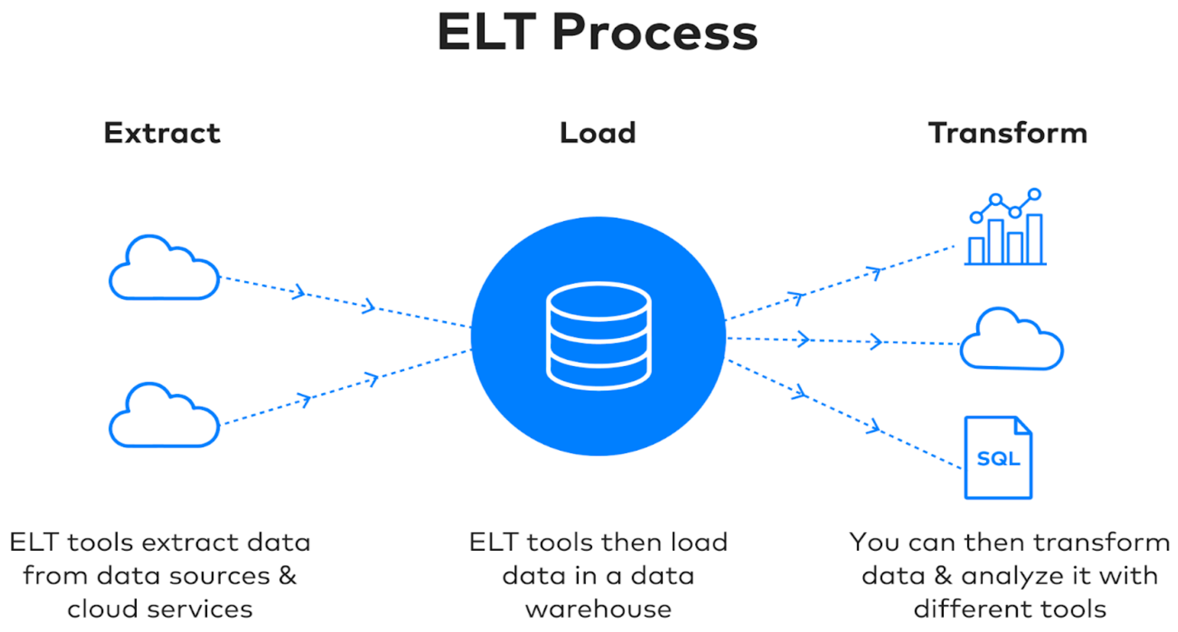
ETL

- Extract, Transform, Load is the process of combining data from multiple sources into a DWH



ELT

- Extract, Load, Transform leverages the processing power of modern cloud DWH to transform data after loading
- It shifts the work to the warehouse



ETL VS ELT

Extract Transform Load

- Better for structured data and scenarios where immediate transformation is necessary
- Slower, transformation happens BEFORE loading
- Best for smaller, complex data sets
- Higher costs
- Requires custom security solutions

Extract Load Transform

- More suitable for large-scale, unstructured data and cloud-based environments
- Offers flexibility and cost-effectiveness
- Faster, data loaded first and transformed in parallel
- Built-in security features